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1. The solution of  $(e^x + 1)ydy = (y + 1)e^x dx$  is

**NIMCET 2017**

- (a)  $e^x = c(e^x + 1)(y + 1)$   
 (b)  $e^y = e^x + y + 1$   
 (c)  $y = (e^x + 1)(y + 1)$   
 (d) None of the above

2. The solution of the differential equation

$$\frac{dy}{dx} = e^{x+y} + x^2 e^y \text{ is}$$

**NIMCET 2017**

- (a)  $e^{x-y} + \frac{x^3}{3} = C$   
 (b)  $e^x + e^{-y} + \frac{x^3}{3} = C$   
 (c)  $e^x - e^{-y} + \frac{x^3}{3} = C$   
 (d) None of the above

3. Through any point  $(x, y)$  of a curve which passes through the origin, lines are drawn parallel to the coordinate axes. The curve given that divides the rectangle formed by the two lines and the axes into two areas, one of the which is twice the other, represent a family of

**NIMCET 2018**

- (a) circles (b) parabolas  
 (c) hyperbolas (d) straight line

4. The curve satisfying the differential equation  $ydx - (x + 3y^2)dy = 0$  and passing through the point  $(1, 1)$  also passes through the point.....

**NIMCET 2019**

- (a)  $(\frac{1}{4}, \frac{1}{2})$  (b)  $(\frac{1}{4}, -\frac{1}{2})$   
 (c)  $(-\frac{1}{3}, \frac{1}{3})$  (d)  $(\frac{1}{3}, -\frac{1}{3})$

5. Let  $f(x)$  be a polynomial satisfying  $f(0) = 2, f'(0) = 3$  and  $f''(x) = f(x)$ . Then,  $f(4)$  is equal to

**NIMCET 2019**

- (a)  $\frac{5(e^8 - 1)}{2e^4}$  (b)  $\frac{5e^8 - 1}{2e^4}$   
 (c)  $\frac{2e^8}{5e^8 - 1}$  (d)  $\frac{2e^4}{5(e^8 + 1)}$

6. If  $(x) = x^n$ , then  $f(1) + \frac{f'(1)}{1!} + \frac{f''(1)}{2!} + \dots + \frac{f^{(n)}(1)}{n!}$  is equal to: (Differentiation)

- (a) 0 (b)  $2^n$   
 (c)  $2^{n-1}$  (d)  $\frac{n(n+1)}{2}$

**NIMCET 2024**

## ANSWER KEY

|    |   |    |   |    |   |    |   |    |   |
|----|---|----|---|----|---|----|---|----|---|
| 1. | D | 2. | B | 3. | B | 4. | C | 5. | B |
| 6. | B |    |   |    |   |    |   |    |   |

## Analysis 2007 to 2025

### Questions Asked in NIMCET 2007 to 2025

| Years       | No. of Questions | Years       | No. of Questions |
|-------------|------------------|-------------|------------------|
| NIMCET 2007 | 00               | NIMCET 2016 | 00               |
| NIMCET 2008 | 00               | NIMCET 2017 | 02               |
| NIMCET 2009 | 00               | NIMCET 2018 | 01               |
| NIMCET 2010 | 00               | NIMCET 2019 | 02               |
| NIMCET 2011 | 00               | NIMCET 2020 | 00               |
| NIMCET 2012 | 00               | NIMCET 2021 | 00               |
| NIMCET 2013 | 00               | NIMCET 2022 | 00               |
| NIMCET 2014 | 00               | NIMCET 2023 | 00               |
| NIMCET 2015 | 00               | NIMCET 2024 | 01               |
|             |                  | NIMCET 2025 | 00               |

## SOLUTION

1. (d) We have,

$$(e^x + 1)ydy = (y + 1)e^x dx \Rightarrow \frac{y}{y+1} dy = \frac{e^x}{e^x+1} dx$$

On integrating both the sides, we get

$$\int \frac{y}{y+1} dy = \int \frac{e^x}{e^x+1} dx \Rightarrow \int \left(1 - \frac{1}{1+y}\right) dy = \int \frac{e^x}{e^x+1} dx$$

$$\Rightarrow y = \log(y + 1) = \log(e^x + 1) + \log C$$

$$\Rightarrow y = \log(y + 1) + \log(e^x + 1) + \log C$$

$$\Rightarrow y = \log[C(y + 1)(e^x + 1)]$$

$$\Rightarrow e^y = C(y + 1)(e^x + 1)$$

2. (b) Given differential equation is

$$\frac{dy}{dx} = e^{x+y} + x^2 e^y = e^y (x^2 + e^x)$$

$$\Rightarrow \frac{dy}{e^y} = (x^2 + e^x) dx \quad [\text{Using variable separable form}]$$

Integrating both sides, we get

$$\int e^{-y} dy = \int (x^2 + e^x) dx \Rightarrow -e^{-y} = \frac{x^3}{3} + e^x + C$$

$$\Rightarrow e^x + e^{-y} + \frac{x^3}{3} = C \quad [\text{where, } C \text{ is constant}]$$

3. (b) Area of OPB = 2 (area of OAP)

$$\therefore xy - \int_0^x y dx = 2 \int_0^x y dx$$

$$\Rightarrow xy = 3 \int_0^x y dx$$

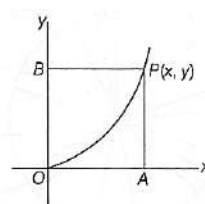
Differentiation with respect to  $x$ , we get

$$y + x \frac{dy}{dx} = 3y \Rightarrow \frac{dy}{y} = 2 \frac{dx}{x} \Rightarrow \log y$$

$$= 2 \log x + \log c \Rightarrow \log y = \log cx^2 \Rightarrow y = cx^2$$

This is an integrating yields  $y = cx^2$ .

Hence, it is a parabola.





4. (c) We have,  $ydx - (x + 3y^2)dy = 0$

$$\Rightarrow ydx - xdy = 3y^2dy \Rightarrow \frac{ydx - xdy}{y^2}$$

$$= 3dy \Rightarrow \int d\left(\frac{x}{y}\right) = 3 \int dy$$

On integrating both sides, we get  $\frac{x}{y} = 3y + C$

This curve is passes through (1,1)

$$\therefore 1 = 3 + C \quad \text{Hence, } C = -2$$

$$\text{Finally curve is } \frac{x}{y} = 3y - 2$$

It passes through  $\left(-\frac{1}{3}, \frac{1}{3}\right)$ .

5. (b)  $f(x)$  be a polynomial satisfying.

$$f(0) = 2, f'(0) = 3, f'(x) = f(x), f(4) = ?$$

$$f''(x) = f(x) \Rightarrow \frac{d^2y}{dx^2} = y$$

Solution of this differential equation is

$$y = Ae^x + Be^{-x}$$

$$f(x) = Ae^x + Be^{-x} \Rightarrow f(0) = A + B$$

$$f'(x) = Ae^x - Be^{-x} \Rightarrow f'(0) = A - B$$

$$A + B = 2$$

$$A - B = 3$$

$$2A = 5$$

$$A = \frac{5}{2}, B = -\frac{1}{2}$$

$$f(x) = \frac{5}{2}e^x - \frac{1}{2}e^{-x}$$

$$f(4) = \frac{5}{2}e^4 - \frac{1}{2}e^{-4} = \frac{5(e^8) - 1}{2e^4}$$

6. (b)  $f(x) = x^n$  then  $f(1) + \frac{f'(1)}{1!} + \frac{f''(1)}{2!} + \dots + \frac{f^n(1)}{n!}$

$$f(x) = x^n \quad f(1) = 1$$

$$f'(x) = nx^{n-1} \quad f'(1) = n$$

$$f''(x) = n(n-1)x^{n-2} \quad f''(1) = n(n-1)$$

$$f^n(x) = n(n-1)(n-2) \dots 1 \quad f^n(1) = n!$$

$$f(1) + \frac{f'(1)}{1!} + \frac{f''(1)}{2!} + \dots$$

$$1 + \frac{n}{1!} + \frac{n(n-1)}{2!} + \frac{n(n-1)(n-2)}{3!} + \dots + \frac{n!}{n!}$$

$${}^nC_0 + {}^nC_1 + {}^nC_2 + \dots + {}^nC_n$$

$$= 2^n$$